



MASENO UNIVERSITY

UNIVERSITY EXAMINATIONS 2015/2016

FIRST YEAR SECOND SEMESTER EXAMINATIONS FOR THE DEGREE OF BACHELOR OF SCIENCE (ACTUARIAL SCIENCE, WITH INFORMATION TECHNOLOGY)

MAIN CAMPUS

MAC 106: MACRO-ECONOMICS

DATE: 12TH July, 2018

TIME: 8.30-11.30am

INSTRUCTIONS

- Answer question ONE and any other TWO questions.
- Show all necessary workings.
- Begin each question on a new page.

QUESTION ONE (COMPULSORY) (30mks)

(a) Define the following concepts:

- (i) Terms of Trade
- (ii) Currency Convertibility
- (iii) Structural unemployment
- (iv) Budget Deficit
- (v) Inflationary Spiral [5mks]

(b) Give and briefly explain five causes of inflation. [5mks]

(c) ~~Discuss the qualities which anything should possess if it is to be used as~~
money. [5mks]

(d) (i) Give and explain limitations of gross national product as a
measure of the economic well being. [4mks]

(ii) Suppose interest rate in a certain bank is 1% per month. Obtain
the present value of having 10 dollars in each of the next two
years. [2mks]

(e) (i) Explain the concept current account. [2mks]

(ii) Describe the concepts multiplier and accelerator effects. [4mks]

(iii) Give and explain three indicators of economic well being of a
country. [3mks]



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(d) (i) Give and explain limitations of gross national product as a measure of the economic well being.

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(ii) Suppose interest rate in a certain bank is 1% per month. Obtain the present value of having 10 dollars in each of the next two years.

[2mks]

(e) (i) Explain the concept current account.

[2mks]

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[4mks]

(iii) Give and explain three indicators of economic well being of a country.

[3mks]

QUESTION TWO (20mks)

- (a) (i) Explain the following concepts and in each case provide examples:
1. Flow
 2. Stock [2mks]
- (ii) Give and explain three sets of flow - stock relations. [6mks]
- (iii) Explain the concept public debt providing factors of influence in its growth in an economy. [2mks]
- (b) (i) Describe the functions of central bank [2mks]
- (ii) Give the roles played by central bank in the development of a country. [2mks]
- (iii) Give four steps a government would put in place in its effort to control inflation. [4mks]
- (iv) Describe the concept inflationary gap. [2mks]

QUESTION THREE (20mks)

- (a) (i) Explain the concepts national income and currency convertibility. [2mks]
- (ii) Explain the process of constructing national income accounts. [6mks]
- (iii) What is the significance of national income accounting? [2mks]
- (b) (i) State Okun's law of output gap and give its relevance to economic growth of your country. [2mks]
- (ii) State the objectives of international monetary Fund explaining why it has not succeeded in meeting some of its objectives. [5mks]
- (iii) What factors may cause real wage to get stuck at a point above the full employment level? [3mks]

QUESTION FOUR (20mks)

- (a) (i) Define the concept inflation and give various types of inflation. [4mks]
- (ii) Give and explain the major causes of inflation. [5mks]
- b) (i) Give and explain five ways through which central bank controls the use of money in the economy. [5mks]
- (ii) Give trade barriers that restrict international trade and which are basically designed to protect home industries. [5mks]
- (iii) Explain the concept Balance of Payments. [1mk]

QUESTION FIVE (20mks)

- (a) Consider a simple economy with only three commodities. The market price of each commodity is $P_1 = \$50$, $P_2 = \$100$ and $P_3 = \$150$. The production and consumption of each good during the year 2000 was $Q_1 = 200\text{kgs}$, $Q_2 = 250\text{kgs}$ and $Q_3 = 100\text{kgs}$. Assume that in 2002 prices rose to $P_1 = \$60$, $P_2 = \$120$ and $P_3 = \$170$ and quantities produced and consumed increased to $Q_1 = 210\text{kgs}$, $Q_2 = 270\text{kgs}$ and $Q_3 = 110\text{kgs}$,
- (i) Calculate the value of nominal GDP and real GDP, using 2000 as the base year.
- (ii) What is inflation as measured by the GDP deflator?
- (iii) What is the real rate of growth of the economy? [6mks]
- (b) Of what significance are foreign reserves to a country? [6mks]
- (c) Explain why budget is considered the focal point of the government's social and economic policy. [8mks]

QUESTION TWO (20mks)

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1. Flow
 2. Stock [2mks]
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**FIRST YEAR SECOND SEMESTER EXAMINATION FOR
THE DEGREE OF BACHELOR OF SCIENCE IN ACTUARIAL
MATHEMATICS WITH INFORMATION TECHNOLOGY**

MAIN CAMPUS

MAC 102: FUNDAMENTALS OF ACTUARIAL MATHEMATICS

Date: 10th July, 2018

Time: 3.30 - 6.30pm

INSTRUCTIONS:

- Answer Question ONE and any other TWO
- Show all your workings
- Observe neatness and clarity in your workings



QUESTION ONE

- (a) Define the following terms as used in actuarial mathematics
- Morbidity. [3 marks]
 - Homogeneous population. [6 marks]
 - Area Comparability Factor. [1 mark]
- (b) Explain how education influences morbidity. [2 marks]
- (c) Define in words the following single figure indices. [2 marks]
- Crude death rate.
 - Indirectly standardised mortality rate.
 - Standardised mortality ratio.
- (d) You are given the following statistics in respect of a certain country.

Age	Males		Females	
	E_x^c	Mortality rates	E_x^c	Mortality rates
20-29	125000	0.00356	100000	0.00125
30-39	200000	0.00689	250000	0.00265
40-49	100000	0.00989	200000	0.00465
50-59	90000	0.01233	150000	0.00685

Calculate

- Directly standardised mortality rate. [4 marks]
 - Indirectly standardised mortality rates for the female lives using the combined population as the standard population. [6 marks]
- (e) Explain how a change in the stringency of underwriting may give rise to spurious selection. [3 marks]
- (f) Explain why it must be necessary to subdivide policies by method of selling in order to obtain homogeneous lapse rates. [3 marks]

QUESTION TWO

- (a) Discuss each of the following terms and give an example of each.
- Class selection.
 - Selective decrement.
 - Spurious selection.
 - Adverse selection.
 - Temporary initial selection
- [10 marks]



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 - Temporary initial selection [10 marks]

- (b) Discuss the causal factors that explain observed differences in mortality and morbidity.

[6 marks]

- (c) Explain why lives are subdivided into separate groups for the analysis of mortality.

[4 marks]

QUESTION THREE

- (a) Explain how an insurance company uses risk classification to control the profitability of its life insurance business.

[3 marks]

- (b) Define the following terms giving formulae and defining all notation used

i. Crude death rate.

[2 marks]

ii. Indirectly standardised mortality rate.

[2 marks]

- (c) The following data is extracted from a population census

	All professions		Profession A	
Age band	Population	Deaths	Population	Deaths
20-29	120000	256	12500	30
30-39	178000	458	15000	40
40-49	156000	502	16000	50
50-61	123000	600	14000	60

- i. Calculate the ACF for profession A using the data for All Professions as the standard population.

- ii. Hence or otherwise calculate the indirectly standardised mortality rate and the directly standardised mortality rate.

[8 marks]

- (d) Explain how nutrition affects mortality and morbidity.

[5 marks]

QUESTION FOUR

- (a) Discuss the factors that affect the stated rate of mortality for different occupations. You should consider the factors that affect both the real and apparent levels of mortality.

[15 marks]

- (b) Describe how selection can arise in pension schemes.

[5 marks]

QUESTION FIVE

- (a) Explain how geographical location can affect mortality.

[5 marks]

- (b) The following data has been extracted from the census for the whole of a country with a developed economy and for two of its administrative regions.

Age-group	Region X		Region Y		Country	
	E_x^c	θ_x	E_x^c	θ_x	E_x^c in 000s	θ_x
0-14	590000	136	408000	108	102000	2550
15-39	980000	820	510000	441	16800	13950
40-59	1050000	5690	520000	2816	12900	70950
60-79	870000	42630	260000	11980	8900	418300
80+	11000	18920	36000	6077	1200	204000

(c) The following table gives a summary of mortality for a particular occupational group compared to the whole population. Calculate

- Crude mortality rate for each region and the whole country. [2 marks]
- Standardised mortality rate and
- Standardised mortality ratio for each region by reference to the country as a whole.

[8 marks]

- Comment on your results.

[2 marks]

(d) Give three different forms of selection that would be expected in a group of lives purchasing immediate annuities with an example of each.

[3 marks]

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**FIRST YEAR SECOND SEMESTER EXAMINATION FOR
THE DEGREE OF BACHELOR OF SCIENCE IN ACTUARIAL
SCIENCE WITH INFORMATION TECHNOLOGY**

MAIN CAMPUS

MAC 104: LINEAR MODELS AND FORECASTING

Date: 12th July, 2018

Time: 12.00 - 3.00pm

INSTRUCTIONS:

- Answer Question ONE and any other TWO



LINEAR MODELS AND FORECASTING

Instruction: answer question 1 and any other two questions

1.

- a) Differentiate the following terms as used in linear modeling
- i) Explanatory and response variables (4 Marks)
 - ii) Correlation and regression (4 Marks)
- b) Define a scatter diagram and explain its significance in correlation (3 Marks)
- c) Find the correlation co-efficient given that
 $\text{COV}(X,Y)=3$, $\sigma_x^2 = 4.5$ and $\sigma_y^2 = 5.5$ (4 Marks)

Comment on the correlation co-efficient and its reliability

- d) Calculate the spearman's Rank correlation from the data below (5 Marks)

Marks in statistics(x)	15	20	35	18	40	60	80
Marks in economics (y)	40	30	50	36	20	10	60

- e) Suppose $\bar{y}=10$, $\sigma_y=2$, $\bar{x}=15$, $\sigma_x=4$ and $r=0.8$, $n=25$
 Find the regression equations of Y and X and X and Y (6 Marks)
- f) The mean height obtained from a random sample of size 100 is 64 cm. The standard deviation of the distribution of the height of the population is known to be 3 cm. Test the statement that the mean height of the population is 67 cm at 5% level of significance (4 Marks)

2.

- a) Using the bivariate data set below

X	1	2	3	4	5
Y	10	8	12	16	20

- i) Define the term Residual and show that the sum of residuals is zero (5 Marks)
- ii) Show the Relationships between total variation, explained variation and unexplained variation (6 Marks)
- iii) Calculate the co-efficient of determination and comment on the result (4 Marks)
- iv) Draw the residual plot and comment on it (5 Marks)

3.

- a) For a least squares line of Y on X
 i.e. $\bar{y} = a_0 + a_1x$ show that (8 marks)
 $a_0 = \bar{y} - a_1\bar{x}$ and



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- a) For a least squares line of Y on X
 i.e. $\bar{y} = a_0 + a_1x$ show that (8 marks)
 $a_0 = \bar{y} - a_1\bar{x}$ and

$$a_1 = \frac{N\sum XY - \sum X \sum Y}{N\sum X^2 - (\sum X)^2}$$

- b) The table below shows the Age (x) and Blood Pressure (y) of same patients taken at random in a given hospital

Age (x)	56	42	72	36	63	47	55	49	38	42	68	60
Blood pressure (y)	147	125	160	118	149	128	150	145	115	140	152	155

- i) Find the regression line of Y on X (10 MARKS)
 ii) Estimate the Blood pressure when the age is 50 years (2 Marks)

4.

- a) Ten competitors in a beauty contest are marked by judges in the following order

1 ST JUDGE	1	5	4	8	9	6	10	7	3
2 ND JUDGE	4	8	7	6	5	9	10	3	2
3 RD JUDGE	6	7	8	1	5	10	9	2	3

Use the rank correlation co-efficient to discuss the pair of judges that have the nearest approach to common tastes in beauty (8 Marks)

- b) The table below shows the number of cars in 10,000 (X) owned by taxi companies and the revenue they generate in billions(Y) in a given year in the USA

COMPOANY	A	B	C	D	E	F
CARS (X)	63	29	20.8	19.1	13.4	8.5
REVENUE (Y)	7	3.9	2.1	2.8	1.4	1.5

- i) Draw a scatter plot and comment on the graph (3 marks)
 ii) Complete the correlation co-efficient and comment on it (5 marks)
 iii) Test the significance of the correlation co-efficient (4 marks)

Use $\alpha=0.05$

5. A researcher collects the following data and determine that there is a significant relationship between the age of a copy machine and its monthly maintenance cost

Machine	A	B	C	D	E	F
AGE (X) YRS	1	2	3	4	5	6
MONTHLY COST (Y) Ksh	62	78	70	90	93	103

Calculate

- i) The regression line of Y on X (6 marks)
 ii) The standard error of estimate (6 marks)
 iii) The 95% prediction interval for the monthly maintenance cost of a machine that is 3 years old (8 marks)

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**SECOND YEAR SECOND SEMESTER EXAMINATIONS FOR THE
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WITH INFORMATION TECHNOLOGY**

MAIN CAMPUS

MAC 206: ACTUARIAL MATHEMATICS I

Date: 6th July, 2018

Time: 8.30 – 11.30 am

INSTRUCTIONS

- Answer question ONE and any other TWO questions.

